

What a statistician can do with BEOPS

A reply to the question relayed through Dara: "What can we, as STAT people, do? Check the intensity of the hexagons over time? What are the ideas there?"

Written 10 September 2026 against the record as it stands that day. Every number below was measured from the files, not remembered; where something cannot be done, the reason is a property of the data and is stated as one.

First, the honest correction: the hexagons have no time in them

The population layer is **one static release**, not a series:

Source	Kontur Population Dataset (2023 release), CC BY 4.0
Geometry	H3 resolution 8 — hexagons of about 0.74 km², roughly 460 m across
Extent held here	2,452 hexagons covering the Belgrade window
People	1,719,722
Per hexagon	median 119 people, 90th percentile 2,434, maximum 14,242
Time dimension	none — a single modelled snapshot

So "the intensity of the hexagons over time" cannot be measured from this layer, and no amount of statistics will recover a time axis that was never sampled. Kontur revises the dataset periodically, so a *release-to-release* comparison is possible in principle, but that measures the model's revisions as much as the city, and two points several years apart is not a time series either.

What the instinct behind the question is right about is the unit. The hexagon is the correct spatial unit for this city — it is small enough to be meaningful, it tiles without a projection argument, and it carries a population count. What varies over time is not the hexagons but the measurements, and the productive move is to use **the hexagons as the frame and the instruments as the signal**: a hexagon-level field, populated where instruments support it, and *left empty where they do not*. Which brings the whole question back to the thing this project is actually about.

The number that falls out immediately, before any modelling

Using the hexagons as they are and the 117 distinct instrument locations this record can read:

people living within	of any instrument	share of Belgrade
500 m	299,470	17.4 %
1 km	682,709	39.7 %
2 km	1,086,552	63.2 %
3 km	1,310,416	76.2 %
5 km	1,515,388	88.1 %

Read the other way: **about 633,000 people — 36.8 % of the city — live more than two kilometres from any instrument whose readings this project is permitted to collect.** That is computable today, needs no further data, and is the strongest single figure the collaboration could produce in an afternoon.

It is also the paper's argument in one line: the gap is not sensing technology, it is which signals are published and on what terms. A statistician turning that into a proper population-weighted coverage curve — with the distance-to-nearest-instrument distribution weighted by people rather than by area — would be contributing the headline result, not an appendix.

The seven things worth doing, in the order I would do them

1. Coverage and exposure, with people as the denominator

What exists. 2,452 hexagons with population; 117 instrument locations; hourly or better readings for air quality, temperature, humidity, pressure and wind.

The method. Population-weighted exposure: for each hour, assign a concentration to each hexagon *only* where an instrument supports it, weight by the hexagon's population, and report the distribution — not just the mean. The output is person-hours at concentration, and a coverage statement beside it.

The trap, and it is the important one. The obvious move is to interpolate a smooth concentration surface across the whole city and integrate it against population. **Do not.** With 31 reference instruments over 1.7 million people, most of that surface would be invention, and it would be invention that looks authoritative because it is smooth. The rule this project runs on is that a value the record does not support is *absent*, not estimated. So the deliverable is an exposure figure for the covered population plus an explicit, quantified statement of who is not covered.

What it yields. A defensible exposure statistic, and — more valuable — a measurement of the uncovered population that no city dashboard publishes because publishing it is against the publisher's interest.

2. Citizen sensors against reference stations

What exists. Two independent networks measuring particulates in the same unit:

network	streams	names used	unit
SEPA reference stations (S146)	31 instruments	PM10, PM2.5	µg/m ³
Sensor.Community citizen sensors (S04)	29 instruments	P1, P2	µg/m ³

and 607 instrument pairs lie within 2 km of each other, 198 within 1 km, 62 within 500 m.

One thing to check before anything else, and it is a good example of the local hazard. P1 and P2 are the Sensor.Community convention for PM10 and PM2.5. That convention is almost certainly right — but **it is nowhere in this record**, and the record's rule is that a mapping which is not written down is an assumption, not a fact. Establishing it properly (from the source's own documentation, captured and hashed like everything else) is the first task, and it is exactly the kind of thing that a statistician would otherwise take for granted and a reviewer would ask about.

The method. Standard and well-precedented: pair each citizen sensor with reference stations within a chosen radius, regress one on the other by hour, and estimate bias, proportional error and the humidity dependence that low-cost optical particle counters are known to have. Bland–Altman for the agreement, not correlation — correlation is the classic mistake here, because two instruments can correlate at 0.95 and disagree by a factor of two.

The trap. Citizen sensors are **self-sited**. Where they are is a choice made by whoever bought one, so they are a non-probability sample of locations, and their spatial distribution is informative — it probably tracks income and interest, not pollution. Any city-wide statement built on them needs that treated explicitly (reweighting toward the population distribution, and reporting the weights).

What it yields. A calibration for Belgrade's citizen network, which is useful to the city and not merely to the paper — and a quantified answer to "can the cheap sensors substitute for the expensive ones", which is the question everyone asks and few answer locally.

3. Separating instrument disagreement from the city itself

This is the one I would most like a statistician to take, because the project has hit its own limit here and says so.

There is already a layer that computes, for every parameter measured by more than one source, how far apart the sources are in the same hour:

parameter	shared hours	published	median gap of the widest pair
temperature	23	yes	2.6 °C (closest pair: 0.6 °C)
humidity	24	yes	5.0 %
wind direction	23	yes	29.8°, worst hour 156.9°
pressure	24	withheld	the two sources publish different units

parameter	shared hours	published	median gap of the widest pair
wind speed	23	withheld	the two sources publish different units

The layer states plainly, in its own output, that this gap "contains instrument difference, siting, and genuine local variation" and that it cannot separate them. **A statistician can.**

The method. Treat the gap as a function of separation distance — a variogram. The intercept at zero distance (the nugget) is instrument-plus-siting error; the rise with distance is real spatial structure. The pair inventory supports it: 6,786 pairs in total, median separation 7.7 km, and critically **62 pairs closer than 500 m**, which is where the nugget is estimated. For wind direction this must be done on the circle — the gaps above are already circular distances, because the first version of the layer subtracted bearings like a line and published a 90th-percentile disagreement of 186°, which is impossible on a compass.

What it yields. The sentence "these two thermometers disagree by 2.6 °C" becomes "0.9 °C of that is the instruments and 1.7 °C is the city" — which is the difference between a caveat and a finding. It also gives every other layer an uncertainty it can carry, and it is the input the exposure work in (1) needs before it can honestly put a number on a hexagon that has no instrument in it.

4. Deciding when the record is allowed to say what is "usual"

What exists. A layer that computes, per station and per hour of day, the median of everything this record has received. It currently publishes **nothing**: 12,810 candidate buckets, zero released, because its threshold is *at least 3 distinct days and at least 5 values* and the record is 44 hours old.

The problem with that threshold is that I chose it by hand. Three days and five values is a plausible-sounding rule with no sampling argument behind it, and it is doing real work: it is the sole reason the layer says nothing.

The method. Two questions a statistician settles properly. What is the sampling distribution of an hourly median at $n = 5$, $n = 10$, $n = 30$, for a quantity with this much autocorrelation — and therefore what interval width does each n buy? And should the estimator borrow strength: a hierarchical model with partial pooling across stations would let a station with three days of record use the city's shared shape, with the pooling weight falling as its own record grows.

The trap, and it is a matter of principle rather than technique. The point of the threshold is that a thin bucket is **absent, never provisional**. A hierarchical model must not become a way to publish a guess with a confident face. The acceptable deliverable is an *interval that is honestly wide*, and a rule for when it is too wide to be worth showing at all.

What it yields. The layer starts saying something days or weeks earlier than it otherwise would, with a stated uncertainty — and the paper gets a properly derived threshold instead of a round number.

5. Is a silent sensor silent at random?

What exists. 100,337 rows, 845 series, 28 sources, and a great deal of unevenness: sources differ by more than an order of magnitude in how many distinct hours they have contributed. Every gap is recorded as a gap — a source that answers "nothing" produces a receipt, which is a different object from a source that was not asked.

The method. Model the probability that a given station reports in a given hour as a function of the hour, the station, the weather, and the value it last reported. If reporting probability depends on the value — sensors that drop out in the cold, a feed that fails under load when the city is busiest — then the missingness is **not** ignorable, and every naive mean computed over this record is biased in a direction that can be estimated.

What it yields. Either "missingness here is ignorable and simple means are fine", which is a licence the project currently does not have, or a bias correction and a warning. Both are worth having, and this is the single most standard piece of statistics on the list — which is precisely why it should be done by someone who does it properly.

6. How late a public measurement is, as a distribution

What exists. A layer that measures, per source, how old a value already was when it arrived:

source	median age on arrival	note
Sensor.Community	3.4 min	over 64,254 rows
RHMZ automatic stations	5.5 min	
NOAA aviation METAR	9.5 min	

source	median age on arrival	note
SEPA air quality	116.9 min on the corrected clock	and 52.5 min on the source's own labels, whose minimum is -67.5 minutes
Parking Servis	no figure	0 of 3,969 rows carry a measurement time at all

That negative minimum is a value that reached this record *before its own measurement window had ended*, according to the label the publisher attached to it.

The method. Treat publication delay as a right-skewed, censored distribution and ask whether it is stationary: does it depend on hour of day, on day of week, on how much the network is carrying? Then turn it into the operational statistic nobody publishes — **expected completeness at time t**: if you read this record at 14:00, what fraction of the 13:00 measurements have arrived, and what is your confidence in a figure computed at that moment?

What it yields. Anyone building anything on Serbian public data feeds needs this curve and cannot get it. It is a small, publishable, genuinely useful result, and it belongs to the statistician who does it.

7. The permission landscape itself, as data

What exists. A published dataset (CC BY 4.0) of 215 reviewed sources with the reviewer's own written reason for each, 16 refusals on their own, and 313 permission captures listed by hash.

The method. Small-n, so honest methods only: cross-tabulate refusal against publisher type (state body, municipal utility, private outlet, international), and — as the record lengthens — survival analysis on time-to-permission. The obvious regression is under-powered at $n = 215$ with 16 events and should be described as descriptive rather than inferential.

The caveat that must be stated first. The dataset says of itself that its source list is incomplete and its gaps are not random: it was assembled by reading robots files and terms pages rather than by reading the statute, and it under-represents official publishers, whose material is least encumbered. Any statistic computed on it inherits that bias, and the honest framing is "this is a measurement of how one pair of researchers found sources", not "this is the permission landscape of Belgrade".

What cannot be done yet, and why

- **Anything with the word *trend* in it.** The record holds 44 hours of continuous history and grows by 24 hours a day. Nothing about seasons, about weekday effects, or about whether the air is getting better can be said, and the baseline layer demonstrates the cost by publishing nothing at all.
- **Anything about the hexagons changing.** One release, no time axis. See the top of this note.
- **Health.** This record does not compare its values to limit values and does not say whether air is safe. That is a deliberate refusal, asserted by a test, and it is not an oversight to be corrected by a collaborator: the instrument inherits somebody else's calibration and somebody else's clock, and a health statement needs neither of those to be borrowed.
- **Any city-wide surface.** See the trap in (1).

What the collaboration would need from this side

Nothing that does not already exist. The record is a set of plain files — line-per-row JSON for the measurements, a hexagon list for the population, a JSON layer each for latency and agreement, and a published CSV dataset for the permission landscape. No database, no API key, no proprietary format. A statistician can be handed a copy and be working within an hour, in R or in anything else.

Two conditions come with it, and they are the project's own:

1. **Missing is never zero, and an unknown is never a permission.** A gap in the record is a recorded gap, and it stays one in any analysis built on it.
2. **A number that the record does not support is not published,** however defensible the model that produced it. Intervals are welcome; confident points over empty ground are not.

The short answer, if only one sentence is wanted

*The hexagons do not change over time — they are a single 2023 population release, and their real statistical value here is as the denominator, not as a signal. The measurements change; the hexagons say how many people each measurement is about. Put those together and the first result is already sitting there: **about 633,000 people in Belgrade, 37 % of the city, live more than two kilometres from any instrument this record is permitted to read.** After that, the most valuable thing a statistician could add is separating instrument disagreement from genuine spatial variation — the record measures the disagreement and states, in its own output, that it cannot decompose it.*